

1. **Experiment:** Lentivirus Packaging and Raji Cell Library Construction
2. **Time:** 2025.12.30-2026.01.18
3. **Member:** Qiwen Jiang, Xinru Yuan, Qichang Liu, Kaili Jin, Haochen Xu, Yue Fu
4. **Materials:** Hek293T cells, pCDH plasmid library containing the 27 first-round *de novo* designs, pCMVR8.74 packaging plasmid, pMD2.G envelope plasmid, MW25000 linear PEI, PBS or complete culture medium, Raji cells, Polybrene, Puromycin, RPMI-1640 medium, and fetal bovine serum.
5. **Method:**

- (1) Preparation of Hek293T packaging cells
- ① Hek293T cells were cultured in 10 cm dishes at 37 °C in a 5% CO₂ incubator. Lentiviral transfection was performed when the cultures reached approximately 75% confluence.
- (2) Lentiviral packaging of the first-round pCDH library
- ① The pooled pCDH library plasmids (obtained by homologous recombination, containing 27 designs), pCMVR8.74, and pMD2.G were mixed at a mass ratio of 3:2:1, with a total DNA amount of 30 μg per 10 cm dish and transfection was performed using MW25000 linear PEI (Hieff Trans® Polyethylenimine Linear) at a PEI:DNA mass ratio of 3:1. The transfection mixture composition per well was as follows:

Table 1. Transfection mixture composition.

Transfection tube	Reagent	Volume/Amount
A	PEI (1 mg/mL)	90 μL
	Antibiotic-free medium	910 μL (to 1 mL total)
	library plasmid	15 μg
	Envelope plasmid (pMD2.G)	5 μg
B	Packaging plasmid (pCMVR8.74)	10 μg
	Antibiotic-free medium	Up to 1 mL

- ② 6 h after transfection, the transfection medium was replaced with fresh complete culture medium.
- (3) Virus supernatant collection
- ① The PEI solution (Tube A) and the plasmid mixture (Tube B) were gently mixed separately, then combined by adding Tube B dropwise into Tube A using a micropipette. The mixture was vortexed briefly and incubated at room temperature for 15 min to allow complex formation. The transfection complexes were then added dropwise to the Hek293T cells, and the plates were returned to the incubator.
- ② Viral supernatant collection: At 48 h and 72 h post-transfection, collect the viral supernatant from Hek293T cells, filter through a 0.45 μm filter, and use immediately or store at -80 °C. The filtered viral supernatant was concentrated by ultracentrifugation at 20,000 × g for 99 min at 4 °C. The viral pellet was resuspended in PBS or complete culture medium, aliquoted, and stored at -80 °C until use.
- (4) Raji Cell Inoculation and Transduction
- ① Collect Raji cells from a culture flask.
- ② Centrifuge at 300 × g for 5 min at room temperature.
- ③ Carefully remove the supernatant and resuspend the cell pellet in an appropriate volume of complete medium (RPMI-1640 with 10% FBS and 1% penicillin-streptomycin).
- ④ The cell density was determined using an automated cell counter, and the suspension was diluted to a final concentration of 1 × 10⁶ cells/mL.
- ⑤ Add the calculated volume of cells to an appropriate volume of complete medium to achieve a final concentration of 1 × 10⁶ cells/mL.
- ⑥ For each well of a 6-well plate (or equivalent dish), 2 mL of this cell suspension (2 × 10⁶ cells) was

dispensed. If using a 10 cm dish, adjust the volume to 10 mL (1×10^7 cells). The plates were incubated at 37 °C in a 5% CO₂ incubator for 24 h prior to infection.

- ⑦ Infection: Raji cells in logarithmic growth phase were exposed to the concentrated lentiviral library at a multiplicity of infection (MOI) of 0.3. Polybrene was added to a final concentration of 8 µg/mL. The cells were subjected to spin infection at 1,500 rpm for 1.5 h at room temperature and then transferred to a 37 °C, 5% CO₂ incubator for continued culture.

(5) Puromycin selection and expansion of the Raji cell library

- ① Puromycin selection was initiated 72 h after lentiviral transduction. The selection concentration had been determined by a preliminary kill-curve experiment in which untransduced Raji cells were exposed to 0, 0.5, 1, 2, or 5 µg/mL Puromycin for 5 days. The lowest concentration that completely eliminated untransduced cells was 1 µg/mL and was therefore used for library selection.
- ② Selection was continued for 5 days, with fresh Puromycin-containing medium replaced every 2 days. After selection, the surviving cell pool was expanded for an additional 7 days until the total cell number reached approximately 1×10^7 for downstream flow-cytometric sorting.

6. Result:

We individually constructed lentiviral expression vectors for 27 candidate protein sequences. After lentivirus packaging and Raji cell infection, stable transduced cell pools were obtained through antibiotic selection, successfully establishing a Raji cell library covering all 27 candidate proteins. This library can be used for subsequent screening and functional validation of the candidate proteins.